EXP - 3 THREE LAYER NEURAL NETWORK FROM SCRATCH

AIM: To design and implement a three layer neural network using python and train it with the back propagation algorithm to recognize handwritten digit from the MNIST dataset using appropriate activation and Loss function

ALGORITHM:

* Import the required Libraries such as numpy, pandas, Matplotlib, Scikit-learn.
* Load the MNIST training dataset from the CSV file
* Seperate the input feature (pixel values) and output lable (digit)
* Normalize the pixel values by dividing them by 255
* Reshape the image into $28 \times 28 \times 1$ format
* Convert the output label into one-hot encoded vector.
* Split the dataset into training and validation sets.
* Create a three layer neural network consisting of Input layer, Hidden layer 1, Hidden layer 2, output lay
* Compile the model : optimizer, Loss function, metrics
* Train the network using the back propagation algorithm for the specific number of epochs
* Load the test dataset and normalize it
* Evaluate the model performance based on Training and validation accuracy.

CODE:

```
Import numpy as np
Import pandas as pd
Import matplotlib.pyplot as plt
from Sklearn.model_selection import train_test_split
from tensorflow.keras.model import sequential

train_data = pd.read_csv('/content/train.csv')
print("Shape of train_data:", train_data.shape)
x = train_data.iloc[:,1:]
y = train_data.iloc[:,0]

x = x.apply(pd.to numeric, errors='coerce')
x = x.fillna(0)

x = x.values/255.0
x = x.reshape(-1,28,28,1)
print("Shape of x:", x.shape)
y = to_categorical(y, num_classes=10)
print("Shape of y:", y.shape)

model = Sequential([Input(shape=(28,28,1)), flatten(),
                    Dense(128, activation='relu'),
                    Dense(128, activation='relu'),
                    Dense(10, activation='soft'),

model compile( optimizer="adam',
               loss='categorical_crossentropy',
               metric=['accuracy'])

model summary()
```

```
history = model.fit(x_train, y_train, epochs = 10,
                    batch_size = 32, validation_data = (x_val, y_val))

test_data = pd.read_csv('/kaggle/input/digit-recognizer/test.csv')

x_test = test_data.values / 255.0
y_test = x_test.reshape(-1, 28, 28, 1)

for i in range(5):
    plt.imshow(x_test[i].reshape(28, 28), cmap='gray')
    plt.title(f"Predicted digit: {predicted_total[i]}")
    plt.axis('off')
    plt.show()
```

RESULT :

The three layer neural network was successfully designed and implemented using Python and tensorflow keras.